

## 5 Amino 1MQ Available for Research Use Only

(SubQ dosing information cannot be extrapolated because animal studies used oral dosing)

5-Amino-1MQ has robust animal evidence for fat loss, body recomposition, and metabolic improvements, but has never been examined in any human trials. Long-term risks from sustained alteration of NNMT, NAD<sup>+</sup> metabolism, and methyl-donor balance remain unknown. Serious risks and absence of human trials make this research compound unsuitable for human use without more evidence.

💀 Possibly unsafe (mechanistic evidence)

★ ★ Probably effective (strong animal signals)

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

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5-Amino-1MQ (5-amino-1-methylquinolinium) is a small-molecule NNMT inhibitor with robust preclinical evidence for improving adipose metabolism and NAD<sup>+</sup> biology; human clinical data are absent, with only human experience is anecdotal from online communities.

The evidence for 5-Amino-1MQ's long-term human safety leans towards **unsafe because mechanistic risks from altering NAD<sup>+</sup>/methylation biology offset limited animal tolerability signals** and human safety data are essentially absent.

The evidence for 5-Amino-1MQ's efficacy for improved adipose energy expenditure and fat loss is strong because **robust animal and mechanistic data strongly support adipose metabolic effects** but no peer-reviewed human RCTs confirm translation.

The evidence for 5-Amino-1MQ's efficacy for increasing intracellular NAD<sup>+</sup> and mitochondrial function is good because **biochemical and animal data show preserved nicotinamide salvage and mitochondrial benefits**, yet human trials are lacking.

### Benefits

- **Improved adipose energy expenditure and fat loss** [animal testing]  
Preclinical mouse models show **reduced white adipose mass and adipocyte size** with NNMT inhibition and measurable body-fat reductions.
- **NNMT inhibition can alter intracellular NAD<sup>+</sup> availability and energy metabolism; downstream effects on sirtuin and mitochondrial pathways are context-dependent** [animal testing; theoretical (mechanistic)]

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- NNMT blockade preserves nicotinamide for the **NAD<sup>+</sup> salvage pathway**, supporting **SIRT1 activation** and mitochondrial biogenesis.
- **Improved insulin sensitivity and glucose metabolism** [*animal testing*]  
Rodent studies report **improved fasting glucose and insulin responses**; small clinic reports suggest modest metabolic improvements in humans.
- **Preservation or improvement of lean mass** [*animal testing; consistent individual reports*]  
Mouse studies and user reports note **maintenance of muscle mass during fat loss** and subjective **increases in stamina**.
- **Potential hepatic benefits (reduced steatosis)** [*animal testing*]  
Preclinical data indicate **reversal of hepatic steatosis** and improved liver enzymes in diet-induced models.

### Indications

- **Obesity and body-recomposition**
- **Metabolic syndrome / insulin resistance**

### Contraindications

- **Pregnancy and breastfeeding**
- **Severe hepatic impairment**
- **Methylation-altering drugs**
- **Psychiatric illness/insomnia**
- **Uncontrolled psychiatric illness or severe insomnia**

### Side effects

- **Gastrointestinal upset (nausea, transient GI symptoms)** - anecdotally reported
- **Sleep disturbance or transient insomnia** - anecdotally reported
- **Transient restlessness or increased thermogenesis (feeling hotter, increased sweating)**
- **Possible alteration of methyl-donor metabolites; direction and clinical significance are uncertain**

### Drivers of Diminished Response and Mitigation Strategies

For **receptor desensitization** the risk is unknow/low-plausability because **5-amino-1MQ** is a small molecule inhibitor of nicotinamide N-methyltransferase (NNMT) that indirectly affects receptor signaling by altering NAD and methylation status, and chronic NNMT inhibition could lead to adaptive changes in receptors and transporters, as suggested by preclinical NNMT literature, although direct receptor internalization data are limited. For **neutralizing antibodies** the risk is negligible because 5-amino-1MQ is a small nonpeptide molecule and immunology

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studies indicate that such compounds rarely elicit specific neutralizing antibodies. **For downstream pathway adaptation the risk is high** because sustained NNMT inhibition and altered NAD and methyl donor pools can drive compensatory changes in gene expression, enzyme levels and metabolic fluxes, as shown in animal models of NNMT inhibition, which can reduce the incremental effect over time. For **system level physiological adaptation** whole body metabolic homeostasis, including adipose tissue and liver responses, can reset in response to chronic NNMT inhibition, potentially blunting clinical benefit, a pattern consistent with broader metabolic drug experience. **Overall likelihood of waning is unknown, plausibly moderate to high.**

Plausible mitigation is to use **limited duration courses**, monitor metabolic markers, and consider **cycling** or dose reduction to minimize adaptive resistance.

### Known Unknown Safety Risks

**Human clinical data for 5-Amino-1MQ (an NNMT inhibitor) is completely absent; the strongest evidence of both benefit and safety signals comes from rodent studies. Serious health risks for humans remain largely unknown but include theoretical risks from altered methylation/NAD+ metabolism and potential off-target toxicity observed or inferable from preclinical work.**

- **Potential hepatotoxicity or altered liver metabolism.** Although some rodent studies reported **improved hepatic steatosis**, NNMT is highly expressed in liver and altering its activity could unpredictably affect hepatic methylation and xenobiotic metabolism; **human hepatic safety is untested.**
- **Disruption of one-carbon/methylation balance (SAM/SAH) with downstream epigenetic effects.** NNMT consumes **SAM**; inhibiting it changes methyl donor flux and could **alter DNA/histone methylation** and homocysteine/SAM ratios, potentially affecting gene regulation long-term. Human clinical safety data on methylation biomarkers are lacking.
- **Perturbation of NAD+ metabolism and sirtuin signaling.** While **preserving NAD+** is a proposed benefit, abrupt modulation could have unanticipated effects on **DNA repair, mitochondrial function, and cell survival** in certain tissues; long-term consequences in humans are unknown.
- **Off-target toxicity and unknown systemic effects (cardiac, neurological, oncologic)** Small-molecule inhibitors can have **off-target interactions**; comprehensive toxicology (chronic, reproductive, carcinogenicity) in humans is not available. Preclinical safety packages are incomplete for human risk assessment.
- **Metabolic compensation or rebound effects after cessation.** Rodent metabolic adaptations and community reports suggest **effects may reverse** and long-term metabolic homeostasis consequences are uncharacterized.

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Animal data are promising for metabolic effects, but human clinical trials are essentially absent; serious human risks remain speculative and could include hepatic, methylation-related, and off-target toxicities.

A search of major public trial registries shows no completed, registered large-scale trials for 5-Amino-1MQ as of this Aug 2026.

### Biological mechanism

5-Amino-1MQ is a **selective, membrane-permeable inhibitor of nicotinamide N-methyltransferase (NNMT)**. NNMT methylates nicotinamide (NAM) using **S-adenosylmethionine (SAM)** to produce 1-methylnicotinamide (1-MNA), removing NAM from the **NAD<sup>+</sup>** salvage pathway and consuming methyl donors. **Inhibiting NNMT can increase availability of NAM and alter SAM/SAH balance**, thereby **increasing NAD<sup>+</sup>** availability and **maintaining methylation capacity**. Elevated NAD<sup>+</sup> **activates sirtuins (notably SIRT1)** and downstream effectors (AMPK, PGC-1α), promoting **mitochondrial biogenesis, fatty-acid oxidation, and a shift from energy storage to energy expenditure in adipocytes**. NNMT inhibition also modulates **polyamine flux and epigenetic methylation states**. These cascades explain the preclinical findings of **reduced adipocyte hypertrophy, increased energy expenditure without reduced food intake, improved insulin sensitivity, and preserved lean mass**.

### Evidence Summary

5-Amino-1MQ (1MQ) has **strong mechanistic and animal support for increasing adipose energy expenditure** and **raising intracellular NAD<sup>+</sup>**, but **human trials are absent**, so efficacy is plausible but not proven, and **long-term safety confidence is absent**.

| Criterion                      | Mechanistic / preclinical                                                                      | Human data                                  | Implication                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Target / mechanism             | NNMT inhibition → preserves nicotinamide for NAD <sup>+</sup> salvage; alters SAM/methylation. | Minimal peer-reviewed human data.           | Strong target engagement in preclinical models; systemic consequences remain incompletely characterized. |
| Adipose/weight effects         | Robust mouse VAT loss, increased energy expenditure.                                           | Small case reports / clinic summaries only. | Plausible translation; human proof lacking.                                                              |
| NAD <sup>+</sup> /mitochondria | Cellular and animal NAD <sup>+</sup> increases,                                                | No robust human RCTs.                       | Mechanistically strong; human confirmation needed.                                                       |

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|---------------|------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------|
|               | improved mitochondrial markers.                                              |                                     |                                                       |
| <b>Safety</b> | Animal tolerability reported; mechanism alters methylation (plausible risk). | Human long-term safety data absent. | Potential systemic/epigenetic risks; monitor closely. |

### Conclusions

As of April 2024, there is a **complete lack of human safety data** and some mechanistic possibilities for severe long-term consequences, hinted at in animal studies, suggest this therapy is **only for adventurous, risk-seeking subjects**.

### References

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